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Generative AI for Predictive Maintenance for Manufacturing: A Survey

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Abstract

Recent advances in generative artificial intelligence (GenAI)—particularly large language models, generative adversarial networks, and diffusion models—are reshaping the landscape of predictive maintenance (PdM). These technologies offer new capabilities in synthetic data generation, intuitive human-AI interaction, and multi-modal data analysis, thereby addressing long-standing challenges such as data scarcity and enabling multimodal reasoning across sensor, image, and textual data. While efforts to apply GenAI in PdM are emerging, a comprehensive understanding of its unique strengths and synergies with existing methods, such as AI and knowledge systems, remains lacking. This survey systematically reviews the application of GenAI in PdM, highlighting the emerging roles of image and language generative models, evaluating their impacts in PdM, and proposing future directions toward building trustworthy and real-time PdM solutions.

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1. Introduction

The advent of advanced manufacturing technologies, central to Industry 4.0, has enabled the real-time availability of heterogeneous data from diverse equipment across manufacturing facilities [1]. These continuously flowing data streams provide critical insights into manufacturing processes and equipment [2]. Unlike traditional reactive or periodic maintenance methods—which often fail to detect emergent faults—predictive maintenance (PdM) leverages sensor networks and data analytics to forecast failures and proactively schedule interventions [3]. PdM has shown considerable promise in tasks such as remaining useful life (RUL) prediction [4] and fault identification [5], thereby enhancing the effectiveness and precision of maintenance planning [6]. Empirical studies indicate that implementing PdM can reduce maintenance costs by approximately 10–20% and decrease equipment downtime by 30–40%

[7]. Furthermore, productivity within maintenance planning and scheduling departments has been reported to increase by 30% with the adoption of PdM [8].

Artificial intelligence (AI) holds significant promise in PdM due to its ability to detect subtle patterns and anomalies within sensor data, and it has been the focus of extensive research efforts. Wang et al. [9] studied how deep learning supports various analytics levels, from diagnostics to prescriptive actions. Systematic reviews such as [10] and [11] showcased machine learning's effectiveness in PdM while noting challenges in model selection and integration. Broad overviews of AI-based PdM are provided by [12] and [13], exploring emerging trends like digital twins and ethical considerations. Domain-specific reviews in automotive [14], oil and gas [15], and power [16] demonstrate AI's broader industrial applications in fault detection and maintenance optimization. Across applications, the literature agrees that while AI enhances PdM capabilities, data limitations, particularly for faulty signal samples, and implementation complexity remain key challenges [12, 17].

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Recent advances in generative AI (GenAI), especially large language models (LLMs), have introduced foundational capabilities—such as reasoning over complex and unstructured information, and interacting naturally with human users—that are aligned with the core challenges within PdM. For instance, the ability to generate high-fidelity synthetic data helps mitigate limited failure data and class imbalance in training datasets, enabling more robust model development [18, 19, 20]. Moreover, GenAI's advanced reasoning supports the integration and interpretation of heterogeneous sources—such as sensor data, maintenance logs, and operator notes—allowing for more contextualized decision support [21, 22]. Furthermore, its capacity to leverage and understand knowledge graphs (KGs) supports seamless integration with existing domain knowledge, enhancing the precision and reliability of predictive insights [7, 23]. Critically, GenAI enables natural interactions with human workers through intuitive, conversational interfaces for querying system status, referencing technical documents, explaining predicted failures, or generating step-by-step instructions, thereby fostering transparency and trust [21, 24].

Recent research has also examined the application of GenAI to enhance capabilities in PdM. Khan et al. [25] conducted a systematic review of GenAI in aviation, identifying common use of generative adversarial networks (GANs), variational autoencoders (VAEs) and diffusion models (DMs) to generate synthetic failure data where run-to-failure data is limited. Liu et al. [26] reviewed data augmentation techniques for prognostics and health management. Li [27, 28, 29] proposed Generative Manufacturing Systems that leverage multimodal GenAI to enable human-centered decision-making given disruptions. Shafiee [30] discussed the broader role of GenAI including maintenance optimization. Mikolajewska et al. [31] highlighted the integration of GenAI with digital twins for real-time fault diagnosis and PdM. However, existing efforts mainly focus on a specific aspect of GenAI for PdM or a specific industrial domain. A comprehensive understanding of how GenAI can address persistent challenges in current PdM—such as limited interpretability and implementation complexity—remains underdeveloped. Furthermore, systematic investigation is needed to explore how different GenAI approaches can be effectively complemented and integrated in PdM. This paper advances the state of the art by making the following contributions:

- Analyze the characteristics, mathematical formulations, and complementary strengths of AI and GenAI in PdM.
- Examine image and language generative models in PdM and identify their synergies with established approaches such as time-series analysis, SQL and knowledge graphs.
- Identify the foundational capabilities of GenAI, including LLMs, GAN, VAE and DMs, that advances PdM.
- Evaluate the potential impacts for implementing GenAI in PdM, grounded in empirical evidence.

The remainder of this paper is organized as follows: Section 2 provides a mathematical comparison between conventional AI and GenAI in the context of PdM; Section 3 outlines potential implementation impacts for GenAI; and Section 4 summarizes the key findings and proposes future research directions.

2. AI vs GenAI

Traditional AI methods often formulate PdM as a discriminative/supervised learning task. Let $x \in \mathbb{R}^d$ represent the sensor data (e.g., vibration signals, temperature, or pressure), and y denote the target variable (e.g., remaining useful life $y \in \mathbb{R}$ or a fault label $y \in \{0, 1\}$). A discriminative model f_θ is trained to map x to y by minimizing a loss function over a labeled dataset:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f_\theta(x_i), y_i), \quad (1)$$

where $\mathcal{L}$ is the loss function, typically the mean squared error (MSE) for regression or cross-entropy for classification; θ represents the learnable parameters of the model, and θ^* denotes the optimized parameters obtained by minimizing the loss. Once trained, the model infers outcomes using $\hat{y} = f_\theta(x)$, enabling PdM through anomaly detection or RUL estimation.

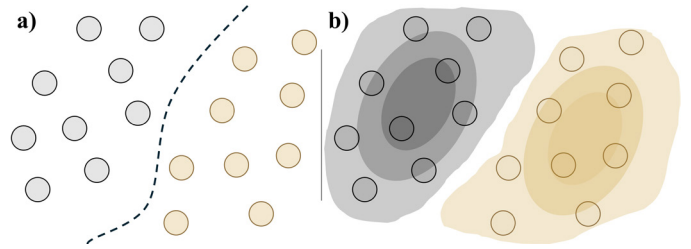


Fig. 1: Comparison between a) discriminative and b) generative learning.

GenAI introduces a paradigm shift by modeling the underlying data distribution $p_\theta(x)$, or the joint distribution $p_\theta(x, y)$, instead of learning a boundary, as shown in Figure 1. The model is trained by maximizing the log-likelihood of the observed data:

$$\theta^* = \arg \max_{\theta} \frac{1}{N} \sum_{i=1}^N \log p_\theta(x_i). \quad (2)$$

GenAI aims to model the underlying data distribution $p_\theta(x)$. This enables GenAI to synthesize data $x^{(s)} \sim p_\theta$ through conditional data sampling. Two common generative approaches include (1) next-token prediction, used in autoregressive models within LLMs, and (2) image generative models, which generate samples in a single pass, namely:

$$\underbrace{x_{t+1} \sim p_\theta(x_{t+1} | x_{1:t})}_{\text{autoregressive models}} \quad \text{and} \quad \underbrace{x \sim p_\theta(z), \quad z \sim \mathcal{N}(0, I)}_{\text{direct sampling (e.g., GANs)}}$$

where z is the latent variable. Based on the prediction and training mechanisms, LLMs and image generative models offer complementary strengths in the context of PdM. LLMs, grounded in autoregressive modeling, provide strong reasoning capabilities and enable intuitive, language-based interpretations of sensor signals. However, their sequential nature and limited context windows can hinder the modeling of global patterns. In contrast, image generative models—trained to map latent variables to data distributions—effectively capture and generate both local and global structures in a single pass, enabling holistic distribution learning and efficient data synthesis, though the reliance on latent variables may reduce the interpretability.

3. GenAI in PdM

GenAI envisions a future PdM paradigm where human operators, databases, and complex systems are seamlessly connected, sensor data and maintenance records are transformed into interpretable, actionable insights, and diverse high-fidelity failure scenarios are generated to overcome data scarcity. We outline GenAI impacts on PdM (Figure 2): (a) *Collaborative Diagnosis* uses LLMs to facilitate human-machine interactions; (b) *Instruction Generation* utilizes LLMs to produce task-specific maintenance guidance; (c) *KG-Augmented PdM* integrates GenAI with domain-specific KGs to apply existing knowledge bases in PdM; and (d) *Data Augmentation* employs image generative models to synthesize diverse, high-fidelity failure data. These impacts are not strictly independent; rather, this classification reflects conceptual distinctions in implementation focus rather than rigid operational separability.

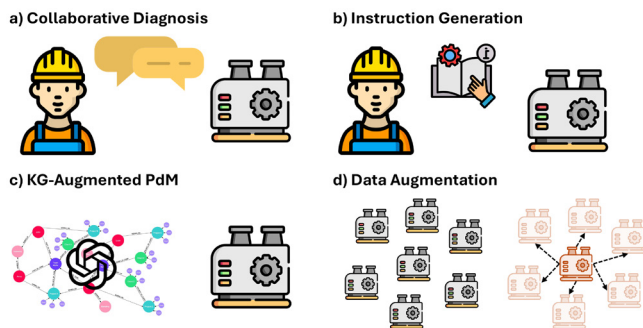


Fig. 2: Envisioned Impacts of GenAI on PdM.

3.1. Collaborative Diagnosis

GenAI, particularly LLMs, enables human-AI collaborative diagnosis through: 1) **contextual understanding**, which enables information retrieval—often via Retrieval Augmented Generation (RAG)—allowing operators to efficiently access relevant data; 2) **reasoning**, which allows the interpretation of both sensor signals and human inputs to directly suggest the potential root causes and diagnostic outcomes.

Recent studies have used LLMs to enhance information retrieval. Kiangala and Wang [32] developed a customized chatbot tool using *GPT-3.5* and *LangChain* to enable users to retrieve factory-monitored data through natural language prompts, reducing troubleshooting time. Gautam et al. [33] introduced a multi-agent LLM framework comprising a supervisor, machine-expertise, visualization, and diagnostic agents, which continuously monitor sensor data and interact with users to provide timely fault diagnosis. Jeon et al. [34] proposed *ChatCNC*, a conversational LLM-based framework that interfaces directly with CNC machine databases. By translating user queries into *SQL* and generating context-aware responses, *ChatCNC* simplifies data access and decision support for operators without requiring technical expertise in data systems.

Other studies tend to utilize generative capabilities to directly identify the potential root cause or diagnostic outcomes.

Qaid et al. [35] proposed *FD-LLM* which reformulates fault diagnosis as a multi-class classification problem using LLMs. They evaluated four open-source LLMs by encoding vibration signals either as text through tokenization or as statistical summaries from time and frequency domains. Alsaif et al. [36] introduced a multimodal diagnostic workflow combining a custom GPT model and *Llama 2*, enabling fault code interpretation through document retrieval, question refinement, and fine-tuned response generation. Xia et al. [37] explored the use of fine-tuned LLMs for diagnosing complex, coupled faults in spacecraft systems and demonstrated how prompt engineering can improve multi-label diagnostic accuracy. Zhao et al. [38] fine-tuned the *Llama3* model for anomaly detection in industrial control systems using behavioral patterns, such as traffic periodicity and communication endpoints, achieving 99.8% accuracy and delivering natural-language explanations of anomalies.

3.2. Instruction generation

Traditional maintenance instructions have primarily depended on rule-based systems, expert-curated knowledge bases, and manual authoring. In contrast, recent advances in LLMs, particularly those incorporating RAG and fine-tuning techniques, enable the dynamic and context-aware instruction generation tailored to the complexity and variability of maintenance tasks and the personalized needs of human operators.

Several studies have explored leveraging RAG to access domain knowledge and customize PdM guidance. For instance, Nagy et al. [39] evaluated eight off-the-shelf LLMs with RAG to improve maintenance instruction generation, evaluated in BLEU and METEOR. Hsu [40] addressed hallucination issues by modeling the relationships among queries, retrieved data, and LLM responses, enhancing the trustworthiness and efficiency of generated instructions. Shi et al. [41] improved the generation of instructions using GraphRAG and chain-of-thought prompting, demonstrating gains in generation structure and accuracy. Similarly, Kaltenpoth and Muller [42] built a proof-of-concept system for generating the instructions based on the documents of the German electricity grid, employing RAG pipelines and LLM-based automatic evaluation strategies.

A set of recent studies incorporate fine-tuning techniques to enhance domain adaptability and reasoning of LLM generations. Angelopoulos et al. [43] fine-tuned LLMs on unstructured maintenance manuals and historical data to enhance technician support and system adaptability. Tao et al. [44] introduced the LORA-KR loss method to fine-tune LLMs on mixed maintenance data, improving reasoning accuracy and reducing hallucinations when generating maintenance schemes. Tao et al. [45] fine-tuned LLMs on a curated dataset of 1,800 QA pairs across aircraft, train, and ship domains, demonstrating improved contextual alignment and instruction generation. Du et al. [46] proposed the *LLM-MANUF* framework, which extends fine-tuning to a comprehensive dataset of manufacturing product equipment operation and maintenance texts to reduce biased decision-making. This framework was validated on a real-world automotive use case, achieving BLEU-4 scores comparable to those of a 30-billion-parameter model.

3.3. KG-Augmented Diagnosis

KGs are commonly applied to represent equipment components, failure modes, sensor relationships, and maintenance procedures. This structured domain knowledge allows LLMs to ground their outputs in verified domain knowledge and also model the causal associations [47]. These approaches leverage the complementary strengths of LLMs' contextual understanding and KGs' rich domain information, resulting in enhanced fault diagnosis accuracy and the interpretability of root-cause analysis. Studies have leveraged LLMs for fault diagnosis and reasoning by incorporating existing KGs to enhance model performance. Wang et al. [48] fine-tuned GPT-3.5 using aircraft maintenance logs enriched with ontology-based hierarchical component structures, which guided the model to more accurately identify defective components and recommend consistent maintenance actions. Bengtsson et al. [49] integrated domain-specific ontologies to extract insights from free-text failure reports, enabling real-time fault analysis and knowledge sharing.

Several studies have used LLMs to construct new KGs to enhance fault diagnosis. Liao et al. [50] proposed an LLM-assisted approach to build a fine-grained fault diagnosis KG for industrial robots by handling nested entities in maintenance texts. Similarly, Wen et al. [51] constructed an *Enhanced Causal Fault Knowledge Graph* that merges theory- and scenario-based knowledge to support interactive troubleshooting. Wu et al. [52] built a KG using *BERT-Transformer-CRF* and LLM-based reasoning for improved machining troubleshooting accuracy. Ma et al. [53] introduced a diagnostic pipeline, *FDRKG-LLM*, which uses LLMs both for KG construction and for reasoning to reduce hallucinations. Likewise, Liu et al. [54] embedded aviation-domain KGs into LLMs via a subgraph-based prefix-tuning approach, achieving high response adoption rates in fault localization tasks. Extended traditional KGs, *Knowledge Hypergraphs (KHG)* were proposed [55, 56] by using schema- and instance-level hyperedges to model complex relationships within fault diagnosis reports.

3.4. Data Augmentation

Machines predominantly operate under normal conditions, resulting in datasets that are heavily skewed toward normal signals. This class imbalance poses a significant challenge for applying traditional AI methods to PdM tasks. GenAI, especially the image generative models, with its ability to learn underlying local/global patterns and data distributions, including scarce fault data, offers a promising avenue to mitigate this limitation.

VAEs have been widely adopted to address the challenge of imbalanced or insufficient fault data in PdM. Liu et al. [57] proposed an interpretable *Adversarial VAE with Sequential Attention (AAVE-SQA)* to augment imbalanced data for rolling bearing fault diagnosis. Zhao et al. [58] utilized a standard VAE to generate synthetic vibration signals for data amplification to enhance CNN-based fault classifiers. To improve the fidelity of generated data, Zhao et al. [59] introduced a *modified Wasserstein Autoencoder (MWAE)* incorporating sliced Wasserstein distance and a squeeze-and-excitation mechanism to ex-

tract better features from limited samples. Wang et al. [60] proposed a *Conditional VAE-GAN* model tailored for gearbox fault diagnosis, enabling realistic fault sample generation across different operating conditions.

GANs have emerged as a powerful tool for generating high-quality synthetic fault data. Deeluea et al. [61] applied GANs to generate abnormal acceleration signals for real-time PdM on low-frequency sensors in injection molding machines. Gao et al. [62] adopted *Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP)* to improve diagnostic accuracy across three industrial datasets by enriching limited fault data. Studies utilized GANs—standard and ACGAN, respectively—to generate vibration signals and balance imbalanced datasets, improving performance on rolling bearing and motor datasets [63, 64]. Hong and Kim [65] combined GAN with VAE and 1D CNN for signal tracking and diagnosis under system disturbances. Other studies like *Flow-based Meta-GAN (GFM-GAN)* by [66] and *improved dynamic time warping distance (MIC-IDTW)* by [67] introduced specific model architectures to learn temporal features and improve stability.

Recent studies have demonstrated the promising role of diffusion models (DMs) in enhancing PdM. Thompson et al. [68] explored the use of DMs on the *PRONOSTIA* dataset, showing how synthetic fault data can be generated with few historical failure records. Solís-Martín et al. [69] used DMs to generate representative time-series samples and improve feature-level explainability in fault diagnosis through the estimation of meta-attributes. For real-time applications, Prevost et al. [70] proposed a *Denoising Diffusion Implicit Model (DDIM)* to detect subtle degradation without prior system knowledge. Ren et al. [71] introduced *DiffMTS*, a temporally augmented conditional adaptive diffusion model, which captures complex temporal dynamics from multivariate time series.

Table 1: Summary of GenAI in PdM Literature

Reference(s)	Model	Training	Impact
[32]-[35]	LLM	PT	Collaborative Diagnosis
[36]-[38]	LLM	FT	Collaborative Diagnosis
[39]-[43]	LLM	PT	Instruction Generation
[44]-[46]	LLM	FT	Instruction Generation
[47]-[48]	LLM	FT	KG-Augmented Diagnosis
[49]-[54]	LLM	PT	KG-Augmented Diagnosis
[57]-[60]	VAE	TfS	Data Augmentation
[61]-[67]	GAN	TfS	Data Augmentation
[68]-[71]	DM	TfS	Data Augmentation

Note: PT = Prompt tuning, FT = Fine-tuning, TfS = Training from scratch

3.5. Impacts of GenAI in PdM

Table 1 summarizes the application of GenAI in PdM. The results show that LLMs are particularly effective for PdM tasks requiring reasoning and text manipulation to enhance information retrieval, generate domain KGs and context-aware instructions across diverse industrial applications, including, conveyors [32], robotics [50], air compressor [41], gearbox [60], accelerator [66] and control system [38]. Concurrently, image

generative models such as VAEs, GANs, and DMs are primarily utilized for data augmentation to mitigate the class imbalance to augment existing datasets, such as CWRU [58, 65, 69], PRONOSTIA [68], and C-MAPSS [71], and industrial scenarios [61, 67]. Overall, this review envisions a future paradigm of PdM in which human operators, existing knowledge bases, historical databases, and real-time machine signals are seamlessly integrated through intuitive interfaces. GenAI serves as an intelligent intermediary—capturing the domain knowledge from heterogeneous data and human expertise while augmenting the insights from limited historical records, ultimately delivering transparent, timely, and context-aware decision support.

4. Conclusions and Future Work

This paper provides a review of GenAI applications in PdM, highlighting its potential for collaborative diagnosis, instruction generation, KG-augmented diagnosis, and data augmentation. It examines how GenAI addresses key PdM challenges like data scarcity and interpretability, offering complementary strengths with existing PdM methods. Future research could focus on establishing integrated intelligence that combines human expertise with GenAI capabilities, and exploring the multimodal integration of sensor, image, and textual data from different sources. Practical challenges—such as deploying large generative models in resource-constrained environments and the risks of hallucinations in LLMs—also warrant further investigation.

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